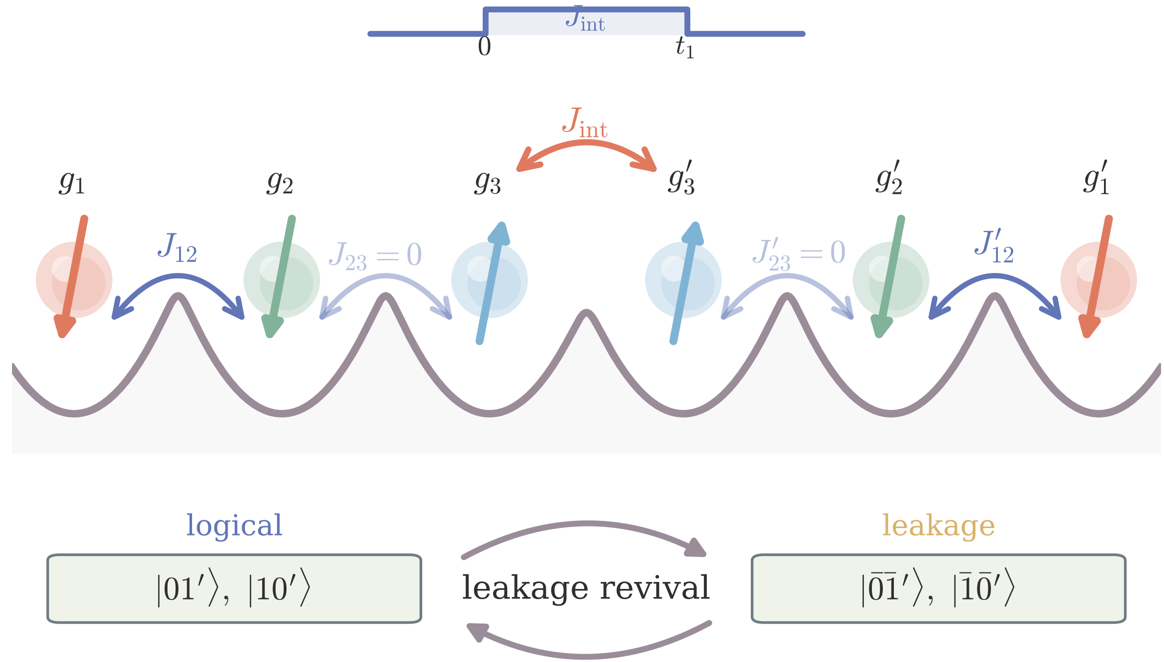

population returns to the logical subspace at t_1

$$U_{\text{log}}(t_1) = \text{diag}\big(e^{-iJ_{\text{int}}t_1/4},\ -e^{i(J_{\text{int}}t_1/4 + \Delta t_1/2)},\ -e^{i(J_{\text{int}}t_1/4 - \Delta t_1/2)},\ e^{-iJ_{\text{int}}t_1/4}\big)$$